Acid and Bases 2

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Ampholytes

Defn: substance that can act either acid or base.

Examples: H_2O, HSO_4^-

Indicators

An indicator is a compound that changes colour according to the pH of the substance. During titrations, the indicator needs to be selected according to the acidity of the salt that will be produced.

Indicator	Colour in acid	Colour in base	Colour at equivalence point	pH range of equivalence
Litmus	Red	Blue		4.5 – 8.3
Methyl orange	Red	Yellow	Orange	3.1 – 4.4
Bromothymol blue	Yellow	Blue	Green	6.0 – 7.6
Phenolphtalein	Colourless	Pink	Pale pink	8.3 – 10.0

Note: Litmus doesn't have a single sharp, well-defined colour at its midpoint.

pH Calculator

- The pH scale runs from 0 (most acidic) to 14 (most basic), measuring how acidic or basic a solution is.*
- pH is calculated from the hydronium ion concentration: * $pH = -\log[H_3O^+]$, directly usable for strong acids/bases, since the $[H_3O^+]$ is known immediately from the given concentration.
- Water ionises itself slightly, even with no acid or base added: * $H_2O(\ell) + H_2O(\ell) \rightarrow H_3O^+(aq) + OH^-(aq)$
  - this is called the auto-ionisation of water and it's the reason any aqeous solution has both H_3O^+ and OH^- present.
- This gives a baseline classification system, all centered on the neutral value * $[H_3O^+] = [OH^-] = 1 \times 10^{-7}\ mol.dm^{-3}$
  - Neutral ~ two concentrations are equal *
  - Acidic ~ $[H_3O^+]$ is higher than 10^{-7} (and higher than $[OH^-]$) *
  - Basic ~ $[H_3O^+]$ is lower than 10^{-7} (and lower than $[OH^-]$) *

Example 1

A few drops of bromothymol blue indicator are added to a potassium hydroxide solution in a beaker. A dilute sulphuric acid solution is now gradually added to this solution until the colour changes (a) Write down the colour change of the indicator.

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Green

Exercise 1

1. Identify the conjugate acid-base pair

1.1 $H_2CO_3 + H_2O \rightarrow H_3O^+ + HCO_3^-$

Base 2 Acid 2

Acid 1 Base 1

1.2 $H_2O + HCO_3^- \rightarrow H_3O^+ + CO_3^{2-}$

Base 2 Acid 2

Acid 1 Base

1.3 $H_2SO_4 + 2H_2O \rightarrow 2H_3O^+ + SO_4^{2-}$

Base 2 Acid 2

Acid 1 Base 1

$A \rightarrow A^-$

Example 2

Calculate the pH of a 0.25 $mol.dm^{-3}$ H_2SO_4 solution: $H_2SO_4(aq) + H_2O(l) \rightleftharpoons 2H_3O^+(aq) + SO_4^{2-}(aq)$

$pH = -\log[H_3O^+]$ ✗c

$\log_b(a)$ ← base

$H_2SO_4 : 2H_3O^+$

$1 : 2$

$0.25 : 0.5$ $(mol.dm^{-3})$

$pH = -\log[H_3O^+] = -\log(0.5) = 0.301$